

Apollo Diagnostics - Laboratory Investigation Report

123, Nungambakkam High Road, Chennai - 600034 | NABL Accredited | Tel: 044-2827-5000

Patient Name:	Vasanth Kumari	Report ID:	APL-META-56-VAS-2026
Age / Gender:	56 Yrs / Female	Category:	Metabolic Syndrome
Weight / Height:	89 kg / 158 cm	BMI:	35.7 kg/m2
Ref. Doctor:	Dr. Arjun Mehta	Report Date:	18-Jan-2026

1. COMPLETE BLOOD COUNT [Circulatory System]

Red Cell Indices

Test	Result	Unit	Reference Range	Flag
Haemoglobin (Hb)	13.1	g/dL	12.0 - 16.0	Normal
Packed Cell Volume (PCV)	39.0	%	36 - 46	Normal
MCV	90.0	fL	83 - 101	Normal
MCH	29.0	pg	27 - 32	Normal
RDW-CV	13.0	%	11.6 - 14.0	Normal

Haematology Comment: Haemoglobin 13.1 g/dL is within normal range. Red cell indices are unremarkable.

White Cells & Platelets

Test	Result	Unit	Reference Range	Flag
Total WBC	7000	cells/cmm	4500 - 11000	Normal
Neutrophils	62	%	40 - 75	Normal
Lymphocytes	28	%	20 - 45	Normal
Platelet Count	2.8	lakh/cmm	1.5 - 4.0	Normal

Comment: WBC and platelet counts within reference limits. No significant leucocytosis, leucopenia or thrombocytopenia.

2. IRON STUDIES [Haematopoietic System]

Test	Result	Unit	Reference Range	Flag
Serum Iron	110	ug/dL	60 - 170	Normal
TIBC	310	ug/dL	250 - 370	Normal
Transferrin Saturation	35.0	%	20 - 50	Normal
Serum Ferritin	80.0	ng/mL	12 - 150	Normal

Iron Studies Comment: Iron stores and transport markers are within normal limits. No iron deficiency or iron overload.

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3. HEPATIC FUNCTION TEST [Hepatobiliary System]

Liver Enzymes

Test	Result	Unit	Reference Range	Flag
SGPT / ALT	100	U/L	7 - 56	HIGH
SGOT / AST	80	U/L	10 - 40	HIGH
Alkaline Phosphatase (ALP)	70	U/L	44 - 147	Normal
Total Bilirubin	0.6	mg/dL	0.2 - 1.2	Normal
Albumin	4.2	g/dL	3.5 - 5.0	Normal

Hepatologist Note: ALT 100 U/L and AST 80 U/L are elevated indicating hepatocellular injury. Likely causes include NAFLD, alcohol use, or viral hepatitis. Liver ultrasound and viral hepatitis screen recommended.

4. RENAL FUNCTION TEST [Urinary System]

Test	Result	Unit	Reference Range	Flag
Serum Creatinine	1.1	mg/dL	0.6 - 1.1	HIGH
Blood Urea Nitrogen	14	mg/dL	7 - 20	Normal
eGFR (CKD-EPI)	64	mL/min/1.73m2	60 - 120	Normal
Serum Uric Acid	4.8	mg/dL	2.4 - 6.0	Normal
Sodium (Na+)	139	mEq/L	135 - 145	Normal
Potassium (K+)	4.0	mEq/L	3.5 - 5.1	Normal

Nephrologist Note: eGFR 64 mL/min/1.73m2 (reduced) and creatinine 1.1 mg/dL (elevated). Consistent with early to moderate chronic kidney disease. Monitor BP, restrict protein, avoid nephrotoxic drugs. Nephrology referral advised if eGFR <45.

5. LIPID PROFILE [Cardiovascular Risk]

Test	Result	Unit	Reference Range	Flag
Total Cholesterol	275	mg/dL	0 - 200	HIGH
HDL Cholesterol	34	mg/dL	50 - 999	LOW
LDL Cholesterol	153	mg/dL	0 - 100	HIGH
Triglycerides	326	mg/dL	0 - 150	HIGH

Cardiologist Note: LDL 153 mg/dL (target <100 mg/dL) and Total Cholesterol 275 mg/dL are elevated. Triglycerides 326 mg/dL. Increased cardiovascular risk. RECOMMENDATION: Mediterranean diet, aerobic exercise, statin therapy consideration if >10-year CV risk >7.5%.

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6. THYROID FUNCTION TEST [Endocrine System]

Test	Result	Unit	Reference Range	Flag
TSH	4.2	uIU/mL	0.4 - 4.0	HIGH
Free T3	3.2	pg/mL	2.3 - 4.2	Normal
Free T4	1.1	ng/dL	0.8 - 1.8	Normal

Endocrinologist Comment: TSH: 4.2 uIU/mL is elevated - consistent with hypothyroidism. FT4 1.1 ng/dL is borderline (subclinical hypothyroidism). Thyroid antibodies (Anti-TPO) and ultrasound recommended. Consider levothyroxine therapy.

7. DIABETES SCREENING [Endocrine - Pancreas]

Test	Result	Unit	Reference Range	Flag
Fasting Blood Glucose	162	mg/dL	70 - 100	HIGH
Post-Prandial Glucose (2hr)	223	mg/dL	0 - 140	HIGH
HbA1c	7.9	%	0 - 5.7	HIGH
HOMA-IR	6.5		0 - 2.5	HIGH

Endocrinology - CRITICAL FINDING: HbA1c 7.9% and FBG 162 mg/dL confirm DIABETES MELLITUS TYPE 2. PPG 223 mg/dL. Immediate antidiabetic therapy (Metformin 500-1000 mg) and dietary modification required. Quarterly HbA1c monitoring and annual fundoscopy/nephropathy screening advised.

8. VITAMINS & BONE [Nutritional / Skeletal]

Test	Result	Unit	Reference Range	Flag
Vitamin D (25-OH)	12.0	ng/mL	30 - 100	LOW
Vitamin B12	450	pg/mL	211 - 911	Normal
Folate (Serum)	12.0	ng/mL	5.4 - 999	Normal
Calcium (Total)	9.4	mg/dL	8.5 - 10.5	Normal
PTH	32	pg/mL	15 - 65	Normal

Nutritionist/Endocrinologist Comment: Vitamin D: 12 ng/mL - deficient (10-20). PTH at 32 pg/mL is within range. Supplementation: 60,000 IU cholecalciferol weekly x 8 weeks, then 1000-2000 IU/day. Recheck in 3 months.

9. INFLAMMATORY MARKERS [Immune System]

Test	Result	Unit	Reference Range	Flag
ESR	35	mm/hr	0 - 20	HIGH
CRP	8.0	mg/L	0 - 5	HIGH
hs-CRP	3.7	mg/L	0 - 1.0	HIGH

Immunology Comment: CRP 8.0 mg/L and ESR 35 mm/hr are elevated. hs-CRP 3.7 mg/L - high CV risk (>3). Inflammatory markers elevated - correlate with clinical symptoms and autoimmune screen.

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CLINICAL SUMMARY & RECOMMENDATIONS

[CRITICAL]	Diabetes Mellitus Type 2 confirmed: HbA1c=7.9%, FBG=162 mg/dL. Antidiabetic therapy required.
[ELEVATED]	LDL=153 mg/dL (HIGH). Total Cholesterol=275. Dyslipidaemia - dietary + pharmacological management.
[ABNORMAL]	Hypothyroidism: TSH=4.2 uIU/mL. Levothyroxine therapy evaluation required.
[ELEVATED]	Elevated liver enzymes: ALT=100, AST=80. Liver disease workup required.
[DEFICIENT]	Vitamin D=12 ng/mL (LOW). Supplementation required.
[HIGH]	CRP=8.0 mg/L, hs-CRP=3.7 mg/L - systemic inflammation present.
[ACTION]	Next appointment: Dr. Arjun Mehta in 4-8 weeks for follow-up and review of supplementation response.

Report reviewed by: Dr. Anitha Krishnamurthy, MD (Pathology) | Verified by Dr. Arjun Mehta

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